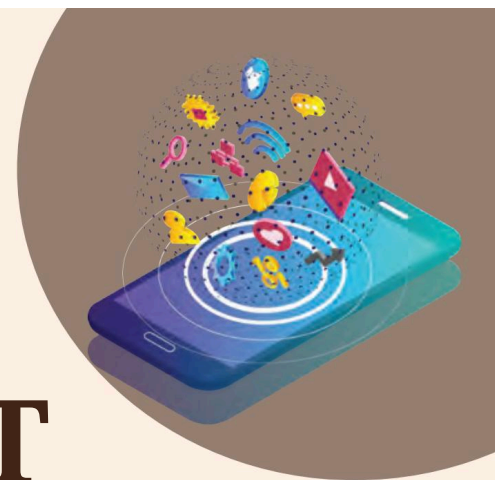


Part-B

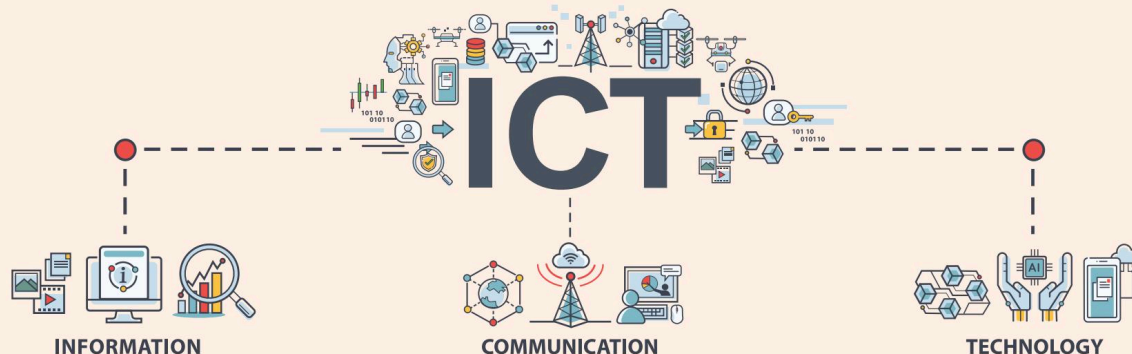
IT Skills

1



Applications of IT

Information and Communications Technology (ICT) is the set of tools and systems that make up modern computing. It generally refers to all devices, networks, applications, and systems that together enable individuals and organisations to communicate or interact in the digital world.



Information Technology

Information Technology (IT) is an essential aspect of modern education. It uses computer systems or digital devices to access, process, and communicate information. It is a combination of both information and technology. It is the common technology used in every business to enhance their business skills. One can use this for both personal and commercial purposes.

Benefits of Information Technology

Following are some of the benefits of IT:

1. Helps in finding potential customers.
2. Helps provide immediate customer service.
3. Online feedback helps in understanding customer requirements.
4. Minimum cost, maximum profit with improved efficiency.
5. Sustainability.

Information Technology-Enabled Services (ITeS)

Information Technology-enabled Services (ITeS) is the process of outsourcing tasks or functions to a third-party business or service provider who uses information technology to provide them. This could be anything from customer service, data entry, tech support, research, or anything else that can be done remotely using technology. India has taken the lead in ITeS, and Indian companies are well-known for outsourcing these services to businesses all over the world. Using technology in this way helps companies become efficient, save money, and focus on what they do best while outsourcing non-essential tasks to specialised service providers.



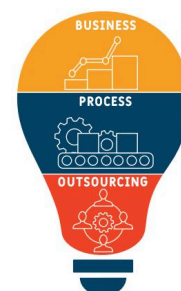
Let us now discuss the most common IT-enabled services that are provided in India.

Call Centres

Call centres are centralised offices set up by companies to handle a large volume of telephone inquiries, requests, or customer support needs. They are staffed with customer service representatives who are trained to assist customers, answer queries, resolve issues, and provide information about products or services. Call centres can handle both inbound calls (incoming calls from customers) and outbound calls (calls made by the company to customers or potential clients for sales, surveys, or follow-ups).

BPO Services

BPO stands for business process outsourcing. It is a business practice in which an organisation hires a third-party service provider or vendor to perform specific business tasks. These tasks can include customer support, technical support, data entry, human resources, accounting, and other non-core functions. BPO services are commonly utilised by businesses to improve efficiency, reduce costs, and access specialised skills without the need for in-house expertise. India is a prominent destination for BPO services due to its large pool of skilled professionals and cost-effectiveness.



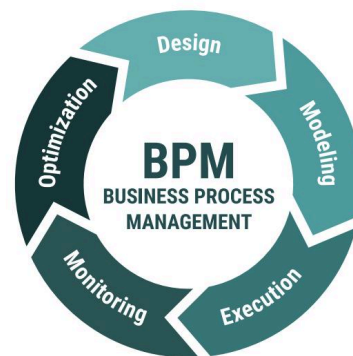
BPO services are often categorised into two main types:

Front Office Outsourcing: This involves customer-related services such as customer support, technical support, sales, and marketing. Companies outsource these tasks to BPO providers to enhance the customer experience and satisfaction.

Back Office Outsourcing: This includes internal business functions such as billing, payroll processing, data entry, and other administrative tasks. Back-office outsourcing helps companies streamline their operations and focus on their core business activities.

BPM Industry in India

The BPM (Business Process Management) industry in India is a significant and rapidly growing sector within the broader ITeS industry. BPM primarily involves the outsourcing of various business processes by organisations worldwide to service providers in India, where skilled professionals handle tasks such as answering phone calls, sorting papers, and performing computer work on behalf of other companies. These tasks are often non-core functions, allowing businesses to focus on their core activities while improving efficiency and reducing costs. The BPM industry in India has indeed experienced substantial growth, contributing significantly to the country's economy.



Some advantages of BPM are as follows:

1. Risks reduction
2. Better organisational control
3. Optimised processes
4. Increased agility
5. Improved collaboration
6. Workflow automation
7. Ensured compliance

Error Alert!

The IT industry primarily involves programming and coding. In reality, IT encompasses a wide range of activities, including hardware, software development, cybersecurity, networking, and more. It's a diverse field with various career opportunities beyond coding.

Structure of the IT-BPM Industry

The IT-BPM industry is an industry that combines information technology and business processes to streamline operations, improve efficiency, and deliver high-quality services. It encompasses a wide range of services, including software development, data management, and customer support.

The IT-BPM industries are categorised based on several key factors:

1. The specific market or region the company serves.
2. The types of services offered by the company.
3. The geographical coverage and reach of the company's services.
4. The extent of the company's operations and the profits it generates.

MNCs: Multinational corporations (MNCs) are global companies with operations in multiple countries. In the IT-BPM industry, they often establish subsidiaries or offices in various countries to tap into local talent and markets.

ISPs: Internet Service Providers (ISPs) provide internet connectivity and services, forming the backbone of the IT-BPM industry by enabling online communication and data transfer.

GIC: Global In-House Centres (GICs) are offshore or nearshore centres established by MNCs to handle specific business functions or IT operations, benefiting from cost savings and access to skilled labour in other countries.

Error Alert!

Some people believe that IT is a static field where knowledge quickly becomes obsolete. However, the IT industry is dynamic and constantly evolving. It is essential for professionals to stay updated with the latest trends and technologies to remain competitive and relevant.

Think and Tell

How has the vibrant startup ecosystem within the Indian IT-BPM industry contributed to its growth and innovation?

Applications of IT

Information technology has a wide range of applications that impact various aspects of our lives and industries. It plays a crucial role in improving efficiency, productivity, communication, and decision-making.

Let's discuss the various applications of IT in different sectors.

IT in Home Computing

Information technology plays a vital role in shaping the landscape of home computing within the ITes sector. With the evolution of technology and changing work dynamics, remote work has become a cornerstone of the ITes industry. IT solutions are the backbone of this transformation, facilitating seamless telecommuting for employees.

IT infrastructure, including secure virtual private networks (VPNs) and remote desktop solutions, enables individuals to work effectively from their homes, ensuring data security with firewalls, antivirus software, and encryption protocols.

The adoption of cloud computing allows easy access to essential tools and data, fostering collaboration and data sharing. Collaboration tools, video conferencing, instant messaging, and project management software are essential for remote teamwork.



IT in Everyday Life

IT is an integral part of everyday life. IT empowers us with digital solutions for various tasks, from online shopping and social networking to mobile banking and telemedicine.

In the ITes industry, IT enables efficient customer service through chatbots and automated systems, streamlines data management, and enhances the overall consumer experience.

IT has an impact on everyday life by making our routines more convenient and businesses more efficient through the automation and digitisation of services and processes.



IT in Library

IT has revolutionised libraries within the ITeS framework. Libraries now employ digital cataloguing, online databases, and electronic resources to provide access to a wealth of information.

IT simplifies the process of searching for and accessing books and materials, making it more efficient for patrons. Additionally, self-checkout systems, automated return terminals, and digital lending platforms enhance the overall library experience.

IT facilitates remote access to digital collections, expanding a library's reach beyond its physical boundaries. In ITeS, IT enables efficient management and maintenance of library systems and supports the broader goal of making information more accessible in an increasingly digital world.

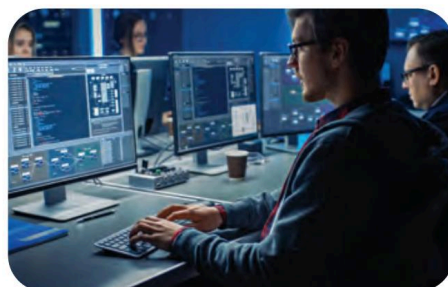


IT at Workplace

IT facilitates seamless communication and collaboration through email, video conferencing, and project management tools. IT systems also manage and secure sensitive data, ensuring compliance with regulations.

It enables remote work, supporting employees with virtual private networks (VPNs) and cloud-based solutions. Monitoring tools track performance and productivity, while data analytics inform strategic decisions.

IT is crucial in the ITeS workplace, streamlining processes, enhancing productivity, and enabling adaptability in an increasingly digital-driven industry.



IT in Education

Let's understand the importance of IT in education as below:

ICT in the Classroom ICT revolutionises education by providing digital tools for teaching and learning. In the classroom, ICT includes interactive whiteboards, educational software, and internet access, enriching the learning experience. It also supports remote learning and collaboration, making education accessible beyond the traditional classroom.

ICT enables educators to create dynamic, engaging lessons and empowers students with resources for research and skill development.

It is a fundamental component of ITeS's contribution to modern education, transforming how knowledge is acquired and shared.



Education Anywhere, Anytime The concept of 'education anywhere, anytime' within the ITeS framework signifies the transformative power of technology in learning. With the advent of digital platforms and online educational resources, individuals can access education on their own terms, breaking free from traditional classroom constraints.

ITeS empowers this shift by offering e-learning platforms, video lectures, and interactive content, enabling students and professionals to acquire knowledge at their convenience.



It fosters lifelong learning, makes education accessible to a global audience, and revolutionises the way we think about acquiring new skills and knowledge, irrespective of the location and time constraints.

Teaching Aids and Media Teaching aids and media in the context of ITeS encompass an array of digital tools and resources that enhance the learning process.

These include interactive multimedia presentations, e-books, video tutorials, and virtual simulations. ITeS leverages these aids to create engaging and effective educational content, making learning more interactive and accessible.

It caters to various learning styles, fosters better understanding, and allows educators to personalise instructions. In ITeS, these aids and media contribute to a more dynamic and versatile learning environment, ensuring that education remains at the forefront of technological advancements.

Learning Management System (LMS) In the ITeS sector, a learning management system (LMS) is a digital platform that streamlines the administration, delivery, and management of educational and training content. An LMS serves as a central hub for course materials, assignments, assessments, and communication between learners and instructors.

ITeS employs an LMS to facilitate e-learning and remote training, making education accessible to a broader audience.

It simplifies tracking progress, offers analytics for performance evaluation, and enhances the overall learning experience by providing a structured, organised, and interactive platform for both learners and educators.

IT in Entertainment

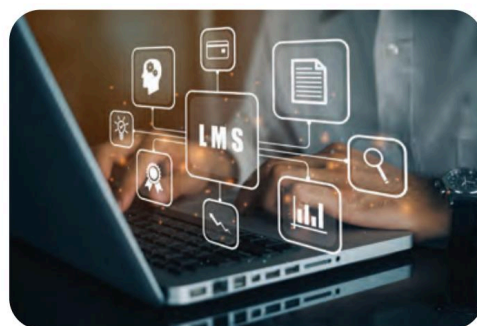
In the ITeS sector, IT has profoundly impacted the entertainment industry. IT-driven innovations have transformed how we consume and create entertainment. Streaming services, digital content distribution, and online gaming have flourished, while virtual reality and augmented reality experiences have pushed the boundaries of immersive entertainment.

IT also plays a critical role in content creation, post-production, and marketing. The ITeS industry facilitates the convergence of technology and entertainment, continuously revolutionising how we enjoy and engage with media, making it more accessible, interactive, and dynamic.

IT in Communication

IT is the backbone of communication within the ITeS sector. It facilitates seamless connectivity and interaction through various channels like email, instant messaging, video conferencing, and social media.

IT supports data management, ensuring the efficient storage and retrieval of information. In ITeS, it's instrumental in customer service through chatbots, helpdesk systems, and data analytics for personalised communication. IT also underpins the development of communication platforms and technologies, contributing to the sector's ability to provide efficient, real-time, and customised services to clients and customers.



IT in Business

IT enables organisations to automate processes, manage data, and enhance decision-making. IT supports everything from data analytics and customer relationship management to financial transactions and supply chain management.

In ITeS, IT solutions enable companies to deliver services efficiently and adapt to the fast-paced, technology-driven business environment.

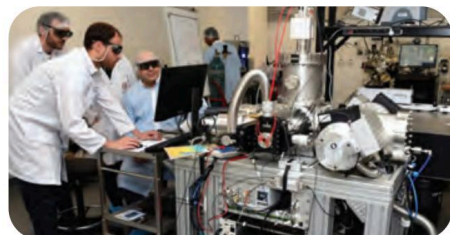
It fosters innovation, streamlines operations, and ensures the security of sensitive information, making it an integral part of achieving success and competitiveness in the modern business landscape.



IT in Science and Engineering

IT facilitates data analysis, simulation, and modelling, enabling researchers and engineers to solve complex problems and develop innovative solutions. It supports scientific research through high-performance computing, making large-scale computations and simulations feasible. In engineering, IT aids in design and prototyping, enhancing efficiency and precision.

It also plays a critical role in data management, helping scientists and engineers organise and analyse massive datasets.

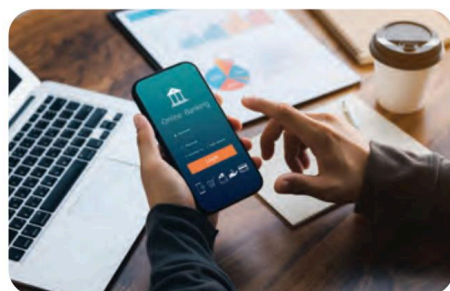


IT in Banking

In the ITeS sector, IT has revolutionised the banking industry. It underpins online and mobile banking, allowing customers to access their accounts, make transactions, and manage finances conveniently.

IT also ensures the security of financial data through encryption and authentication protocols. Additionally, IT supports data analysis for risk management, fraud detection, and personalised customer services.

With technologies like blockchain and fintech innovations, IT in banking is driving efficiency, transparency, and financial inclusion, transforming the sector's operations and the way customers interact with their finances.



IT in Insurance

Information technology (IT) has significantly transformed the insurance industry within the ITeS sector. IT enables insurance companies to streamline their operations through digital underwriting, claims processing, and customer service. Data analytics and machine learning algorithms help assess risk more accurately, enabling better pricing models. Customer engagement and service have improved through online portals and mobile apps. IT also plays a critical role in fraud detection and prevention, safeguarding the integrity of insurance operations. In ITeS, IT is instrumental in enhancing efficiency, reducing costs, and creating a more responsive and customer-centric insurance industry.



IT in Marketing

IT has revolutionised marketing within the ITeS sector. It provides powerful tools for data analysis, targeting, and automation, enabling businesses to reach their target audience with precision.

IT supports customer relationship management (CRM) systems, email marketing, social media analytics, and digital advertising.

In ITeS, IT-driven marketing empowers companies to track consumer behaviour, personalise marketing campaigns, and measure the effectiveness of their strategies.

It has transformed marketing from a mass communication approach to a data-driven, highly targeted, and interactive process, enhancing businesses' ability to connect with customers and drive growth.



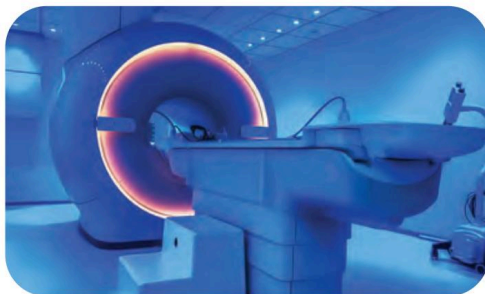
IT in Healthcare

IT has revolutionised healthcare within the ITeS sector. It encompasses electronic health records (EHRs), telemedicine, and digital diagnostics, making patients' information easily accessible and improving care delivery. IT also supports medical research, enabling data analysis and the development of new treatments.

It enhances administrative functions, optimising scheduling, billing, and resource management in healthcare institutions. IT in ITeS empowers healthcare professionals to provide better patient care, improve operational efficiency and drive innovation, ultimately advancing the quality and accessibility of healthcare services.

These sectors rely on IT to improve efficiency, customer experiences, and data management. Use of ICT in diagnosis:

Computerised Axial Tomography Machine A CAT or CT machine is a medical imaging device that uses X-rays and computer processing to generate detailed cross-sectional images of the body's internal structures. These images provide valuable information for diagnosing and monitoring various medical conditions. CT scans are commonly used in hospitals and medical facilities for diagnostic purposes.



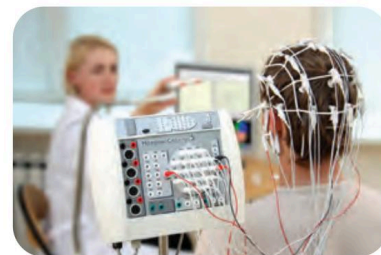
Magnetic Resonance Imaging Machine An MRI is a medical imaging technique that uses strong magnets and radio waves to generate detailed images of the internal structures of the body. This technique provides valuable information for diagnosing a wide range of medical conditions, including neurological disorders, musculoskeletal problems, and various other abnormalities. MRI is a non-invasive and versatile imaging tool widely used in the field of medicine.

Electrocardiogram (ECG) Machine Records the heart's electrical activity, detecting rhythm abnormalities.



Cardiac Screening Machine Assesses heart health, including ECG, echocardiography, and stress tests.

Electroencephalography (EEG) Machine Records brain electrical activity, diagnosing epilepsy, sleep disorders, and brain injuries.



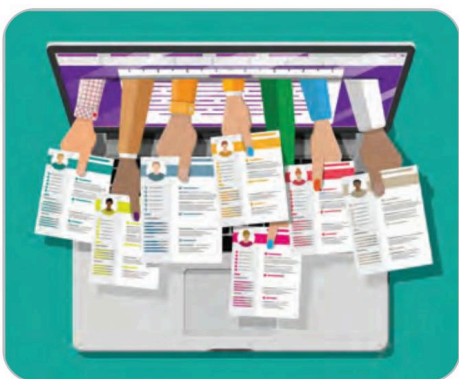
Blood Sugar Testing Machine Measures blood glucose, vital for diabetes management.

Blood Pressure Measuring Machine Measures blood pressure, crucial for cardiovascular assessment.



IT in the Government and Public Service

IT in the government and public services refers to the use of technology, including computers and software, to improve the efficiency, accessibility, and quality of services provided to citizens. By utilising computers, software, and digital tools, governments can streamline various functions such as record-keeping, communication, data analysis, and decision-making. These technologies enable governments to be more transparent, responsive, and efficient in serving the needs of their citizens. IT applications in government contribute to creating a more open, accessible, and responsive public administration system.



Think and Tell

How has the widespread adoption of smartphones and the internet revolutionised the way we access information and interact with technology in our daily lives? Discuss both the positive and negative impacts of this technological shift.

Activity Time

Activity 1: Making a Chart

(Individual Work)

Prepare a chart on the role of IT in everyday life. Include examples of how IT impacts daily activities, such as online shopping, social media, and e-learning.

Activity 2: Preparing Case Study

(Individual Work)

Analyse a real-world case study on the implementation of IT in a business or government service. Identify the benefits and challenges faced by the organisation.

Activity 3: Making a Poster

(Individual Work)

Research and create a poster about various career opportunities in the IT and ITeS industries. Include information on roles such as software developer, data analyst, and IT support specialist.

Chapter Checkup

A Select the correct option.

- 1 What does IT stand for in the context of modern education?
a International Technology **b** Information Technology
c Innovative Techniques **d** Integrated Telecommunications
- 2 Which industry has been significantly transformed by IT, allowing customers to access their accounts, make transactions, and manage finances conveniently?
a Agriculture **b** Manufacturing **c** Banking **d** Construction
- 3 Which of the following are global companies with operations in multiple countries?
a GICs **b** ISPs **c** MNCs **d** None of these

B Fill in the blanks with the most suitable words.

- 1 _____ are staffed with customer service representatives who are trained to assist customers.
- 2 IT also supports _____ research, enabling data analysis and the development of new treatments.
- 3 _____ outsourcing includes internal business functions such as billing, payroll processing, data entry, and other administrative tasks.
- 4 _____ machine records brain electrical activity, diagnosing epilepsy, sleep disorders, and brain injuries.

C State whether the following is *True* or *False*. Correct the statements that are false.

- 1 It is a combination of both information and technology.
- 2 ISPs are offshore or nearshore centres established by MNCs to handle specific business functions, benefitting from cost savings and access to skilled labour in other countries.
- 3 The IT-BPM industry is an industry that combines information technology and business processes to streamline operations, improve efficiency, and deliver high-quality services.
- 4 The adoption of cloud computing in home computing allows easy access to essential tools and data, fostering collaboration and data sharing.

D Answer the following questions. (Solved)

Q1. What are BPO services?

A1. BPO stands for business process outsourcing. It is a business practice in which an organisation hires a third-party service provider or vendor to perform specific business tasks. These tasks can include customer support, technical support, data entry, human resources, accounting, and other non-core functions.

Q2. Mention some advantages of the BPM industry.

A2. Following are some of the advantages of BPM industries:

- Risk reduction
- Better organisational control
- Optimised processes
- Increased agility
- Improved collaboration
- Workflow automation
- Ensured compliance

Q3. Myra wants to know how information technology (IT) has benefited the healthcare industry. Explain this by providing examples of IT applications in the healthcare sector.

A3. Information technology (IT) has revolutionised healthcare within the ITeS sector. IT encompasses electronic health records (EHRs), telemedicine, and digital diagnostics, making patient information easily accessible and improving care delivery. IT also supports medical research, enabling data analysis and developing new treatments.

Answer Key

A 1. b 2. c 3. c

B 1. Call centres 2. Medical 3. Back office 4. EEG Electroencephalography

C 1. True.
2. False. GICS are offshore or nearshore centres established by MNCs to handle specific business functions, benefitting from cost savings and access to skilled labour in other countries.
3. True.
4. True.

Unit Reflection

Key Terms

ITeS: Information technology-enabled services (ITeS) is the process of outsourcing tasks or functions to a third-party business or service provider that uses information technology to provide them.

Call Centres: Call centres are centralised offices set up by companies to handle a large volume of telephone inquiries, requests, or customer support needs.

BPO: BPO stands for Business Process Outsourcing. It is a business practice in which an organisation hires a third-party service provider or vendor to perform specific business tasks.

Front Office Outsourcing: This involves customer-related services such as customer support, technical support, sales, and marketing. Companies outsource these tasks to BPO providers to enhance the customer experience and satisfaction.

Back Office Outsourcing: This includes internal business functions such as billing, payroll processing, data entry, and other administrative tasks. Back-office outsourcing helps companies streamline their operations and focus on their core business activities.

ISPs: Internet Service Providers (ISPs) provide internet connectivity and services, forming the backbone of the IT-BPM industry by enabling online communication and data transfer.

GIC: Global In-House Centers (GIC) are offshore or nearshore centres established by MNCs to handle specific business functions or IT operations, benefiting from cost savings and access to skilled labour in other countries.

Things to Remember

- Information and Communications Technology (ICT) is the set of tools and systems that make up modern computing.
- IT is a common technology used in every business to enhance their business skills. One can use this for both personal and commercial purposes.
- BPO services are commonly utilised by businesses to improve efficiency, reduce costs, and access specialised skills without the need for in-house expertise.
- BPM primarily involves the outsourcing of various business processes by organisations worldwide to service providers in India, where skilled professionals handle tasks, such as answering phone calls, sorting papers, and performing computer work on behalf of other companies.
- The IT-BPM industry is an industry that combines information technology and business processes to streamline operations, improve efficiency, and deliver high-quality services.
- Multinational corporations (MNCs) are global companies with operations in multiple countries.
- IT systems manage and secure sensitive data, ensuring compliance with regulations.
- The concept of “education anywhere, anytime” within the ITeS framework signifies the transformative power of technology in learning.

- IT in communication facilitates seamless connectivity and interaction through various channels like email, instant messaging, video conferencing, and social media.
- IT in banking allows customers to access their accounts, make transactions, and manage finances conveniently.
- IT also supports medical research, enabling data analysis and the development of new treatments.

Test Your Knowledge

A. Select the correct option.

- _____ are centralised set-ups by companies to handle a large volume of telephone inquiries, requests, or customer support needs.
a. Call centres ☐ b. ISP ☐ c. GIC ☐ d. MNC ☐
- _____ are offshore or nearshore centres established by MNCs to handle specific business functions.
a. ISP ☐ b. MNC ☐ c. BPO ☐ d. GIC ☐
- The full form of VPN is _____.
a. Virtual Private Network ☐ b. Virtual Public Network ☐
c. Virtual Personal Network ☐ d. Virtual Package Network ☐
- IT enable _____ companies streamline their operations through digital underwriting, claims processing, and customer service.
a. Banking ☐ b. Marketing ☐ c. Insurance ☐ d. Healthcare ☐

B. Fill in the blanks with the most suitable words.

- ITeS empowers _____ professionals to provide better patient care, improve operational efficiency, and drive innovation.
- The _____ industry in India has indeed experienced substantial growth, contributing significantly to the country's economy.
- Multinational corporations are _____ companies with operations in multiple countries.
- The adoption of _____ allows easy access to essential tools and data, fostering collaboration, and data sharing.

C. State whether the following is *True* or *False*. Correct the statements that are false.

- IT systems manage and secure sensitive data, ensuring compliance with regulations.
- IT in marketing allows customers to access their accounts, make transactions, and manage finances conveniently.
- Information technology plays a vital role in shaping the landscape of home computing within the ITeS sector.
- It is a combination of both information and teaching.

D. Short answer-type questions.

- List any two benefits of IT.
- What are BPOs?

E. Long answer-type questions.

- Differentiate between call centres and BPO services.
- Explain the concept of the BPM industry in India.

F. Competency-based questions.

- Shagun wants to know the key factors by which IT-BPM industries are categorised. Mention them to her.
- Ritik has to explain the use of IT in home computing. Help him to do so.